

# Harsh Sharma

Senior Software Engineer | Backend & Distributed Systems

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## Experience

**IDFC First Bank**, Senior Software Engineer – Bangalore, KA

Dec 2025 – Present

- Shipped the Nominee Management and Audit Service from concept to production in under one week using Java Spring Boot, serving more than 1 million customers from day one.
- Designed and developed an Interest Calculator service in Go capable of handling complex calculations across multiple banking product types while maintaining a complete audit trail.
- Built a Standing Instruction (SI) service from scratch and integrated it synchronously with TCS BaNCS Core Banking System (CBS) to automate recurring banking transactions and payment instructions.
- Owned end-to-end service development including architecture, implementation, integration, deployment, and production readiness in a regulated banking environment.

**Kotak Mahindra Bank**, Software Engineer - 2 – Bangalore, KA

May 2024 – Dec 2025

- Designed and implemented a large-scale report generation pipeline using NumPy and Pandas, reducing processing time from 35–55 days to under 5 minutes while processing more than one million records per batch.
- Designed and developed the Risk-Central platform backend using FastAPI, JWT authentication, and WebSocket-based feedback mechanisms for 300+ monthly users, and built dependency-aware Apache Airflow DAGs across 5+ risk modules to orchestrate batch and real-time data quality validation workflows.
- Led platform engineering initiatives by building a reusable Python SDK adopted across 7+ microservices and establishing Prometheus/Grafana-based observability for 100+ API endpoints, improving developer productivity, SLA visibility, and system performance monitoring.
- Designed dependency-aware Apache Airflow DAGs supporting data quality validation workflows across 5+ risk modules for both batch and real-time processing.
- Automated CI/CD pipelines through Azure DevOps for 20+ services, reducing deployment time by 70% while supporting near-zero downtime releases.

**Juspay**, Software Engineer -Integration – Bangalore, KA

April 2022 - April 2024

- Engineered a real-time event-driven data platform using Rust, Apache Kafka, and rdkafka, processing 5M+ payment events daily with 99.99% delivery accuracy, contributing to the reliability and scalability of high-throughput payment infrastructure.
- Designed ClickHouse materialized views powering real-time analytics dashboards for 10,000+ merchants, delivering transaction, refund, dispute, and payment-method insights with sub-500ms query latency.
- Built a scalable Feature Flag Management Platform using FastAPI and Trie-based data structures, supporting 100,000+ daily evaluations with region-specific and user-specific targeting while enabling zero-downtime configuration updates.
- Enhanced platform performance by integrating Kong API Gateway for service-to-service communication, reducing authentication latency by 65% and improving throughput by 40%, while maintaining Stripe and Bamboo payment integrations supporting approximately 1 cr rupees in monthly transaction volume across five product teams.

**Apisero**, Software Engineer – Remote

May 2021 - March 2022

- Optimized Salesforce scheduler APIs using DataWeave, achieving full synchronization accuracy across more than one million records.
- Developed FHIR-compliant healthcare APIs on MuleSoft CloudHub, improving HL7 interoperability across healthcare systems.

## Education

**Aditya Engineering College**, B.Tech. in Computer Science and Engineering

Aug 2017 – Aug 2021

- GPA: 7.98/10.0

## Projects

**Stock Predictor Web Application**

- Developed a web-based stock price prediction platform using Django for backend services & LSTM (Long Short-Term Memory) neural networks to forecast stock prices based on historical data (last 5 years).

**Space X – 3D Game**

- Designed & engineered a 3D interactive game using Unity Engine, with spacecraft assets modeled in Blender.

## Skills

**Programming Languages:** Rust (Actix), Go (Gin), Python (FastAPI, Flask, Django, Alembic, SQLAlchemy, Pandas, Numpy), Java (Spring boot)

**Cloud Architecture:** Apache Airflow, Kafka, ClickHouse, AWS (EC2, S3, Glue, Lambda, EKS, VPC, Route53), Kong, Redis

**Databases:** PostgreSQL, Clickhouse

**DevOps & CI/CD:** GitHub, Jenkins, Docker, Kubernetes, Azure DevOps.

**Monitoring & Alerting:** Grafana, Loki, OTEL(OpenTelemetry), Promtail, ZooKeeper.